

FINSTANT

PROJECT DELIVERY REPORT

Financial Knowledge Engine & Analytics Platform

Developed at DPL by WHIteWalkers

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1. Executive Summary

Modern organizations depend continuously on financial and operational metrics stored across complex spreadsheets to drive strategic decision-making. However, extracting fast, verified, and contextual answers from these records remains plagued by administrative friction, calculation risks, and severe communication bottlenecks.

FINSTΛNT was engineered and delivered to eliminate this operational divide. It provides an intelligent, conversational knowledge interface designed to sit directly on top of raw multi-sheet Excel spreadsheets (.xlsx). Non-technical managers and leadership simply ask questions in plain English through any desktop or mobile browser. The platform inspects relational schemas, executes verified SQL aggregations, and delivers 100% mathematically accurate calculations, structured tables, and visual charts in under 5 seconds—all inside a secured, private cloud environment running at an ultra-low operational cost of $5 to $8 per month.

This document constitutes the official Project Delivery Report, demonstrating full completion of all project objectives, verifying deliverables against commitments, outlining security and quality assurance benchmarks, and establishing acceptance sign-off.

2. Problem Identified

During comprehensive workflow discovery, the engineering team identified five systemic bottlenecks inherent in traditional spreadsheet management and conventional market software:

- **Rigid Market Tools Fail on Custom Data:** Standard accounting suites (e.g., QuickBooks, Xero) only cater to rigid, generic bookkeeping templates. They fail completely when a business requires tailored analytics on custom column structures, dynamic operational formulas, and unique organizational metrics.
- **Trapped on Office Desktops & Zero Remote Access:** Critical spreadsheets typically reside on local office workstations or internal drives. Decision-makers traveling, attending offsite meetings, or working remotely are paralyzed—unable to retrieve essential financial figures until someone physically goes to the office.
- **Person-to-Person Dependency Bottlenecks:** Access to vital information is locked behind the specific individuals who maintain each workbook. Simple requests stall for hours or days when key personnel are in meetings, on leave, or unavailable, creating unnecessary operational standstills.
- **Over-Sharing & Privacy Vulnerabilities:** When an employee only requires a single figure from one tab, they are routinely emailed the entire master workbook. This needlessly exposes confidential executive payroll, profit margins, and unrelated department records across the organization.
- **Manual Formula Chaos & Public AI Risks:** Manually sifting through thousands of rows invites human calculation error and broken spreadsheet formulas. Conversely, feeding private financial figures into public internet AI tools poses severe compliance violations and data leak risks.

3. Platform Work

FINSTΛNT operates on an automated, multi-tiered pipeline that bridges human conversation with deterministic database operations. The architecture is engineered around four core functional workflows:

3.1 Ingestion & Dynamic ETL Processing

Upon upload of an .xlsx workbook, the ingestion engine extracts each sheet tab, normalizes messy column headers into standardized identifiers, and automatically infers data types (currency, dates, percentages, text). Rather than forcing data into static structures, the engine dynamically constructs relational database tables matching the exact layout of the workbook. Comprehensive metadata profiles (column types, sample values, row counts) are cataloged in real-time, allowing immediate querying without restarting system services.

3.2 Natural Language Understanding & Schema Linking

When an authorized user submits an inquiry in everyday English, the query engine performs semantic schema linking. It maps the user's vocabulary to the exact table names and column headers in the active catalog. The platform maintains multi-turn conversational context, allowing users to ask sequential follow-up questions (e.g., *"What was total payroll in Q2?"* followed by *"Break that down by department"*) without re-stating earlier query parameters.

3.3 Deterministic Execution Engine (Eliminating Hallucinations)

A critical innovation in FINSTΛNT is the strict decoupling of linguistic reasoning from mathematical computation. The language model is used strictly to translate plain English into optimized SQL queries. The mathematical calculations (sums, averages, ratios, rankings) are executed directly by the relational database engine. This architectural separation guarantees 100% computational accuracy and completely prevents mathematical hallucinations.

3.4 Interactive Visualizations & Executive Reporting

Query results are returned through synchronized presentation layers: a natural-language executive summary, dynamic SVG charts (bar graphs, line charts, area charts, and pie charts automatically selected based on data shape), and an interactive data table supporting text search, column sorting, pagination, and one-click PDF and CSV export.

4. Delivery Against Commitment

Every feature, performance milestone, and operational capability promised during project initiation has been completely developed, verified, and delivered into production. The matrix below details the delivery status against commitments:

Committed Scope	Target Objective	Delivered Status
Natural Language Analytics	Allow non-technical users to query spreadsheets using plain English.	**Delivered:** Full natural language querying with sub-5-second turnaround across multi-sheet workbooks.
Custom Spreadsheet Ingestion	Natively handle non-standard spreadsheets and custom business metrics.	**Delivered:** Automated multi-sheet ETL, dynamic table creation, and automated schema cataloging.
Remote Access & Mobility	Eliminate desktop machine dependencies with centralized access.	**Delivered:** Fully responsive cloud deployment accessible via authorized mobile, tablet, and desktop browsers.
Role-Based Access Control	Enforce strict administrative boundaries and data segregation.	**Delivered:** Granular role assignment; standard users can only query approved datasets; full admin controls.
Zero Over-Sharing Privacy	Prevent circulation of sensitive master workbooks.	**Delivered:** System surfaces only the specific answer requested; raw files are never transmitted across endpoints.
Live Dataset Lifecycle	Support updating and purging files without system downtime.	**Delivered:** One-click live file replacement and safe dataset deletion with dynamic table drops.
Deterministic Accuracy	Ensure zero arithmetic errors or hallucinated numbers.	**Delivered:** SQL-driven relational calculations guaranteeing 100% mathematical precision.
Executive Reporting	Provide instant visual analytics and report generation.	**Delivered:** Dynamic Recharts visualizations, interactive data tables, and downloadable PDF reports.
Ultra-Low Operating Cost	Eliminate costly per-user enterprise licensing fees.	**Delivered:** Efficient cloud architecture operating between $5 and $8 per month with zero seat penalties.

5. Testing, Security and Deployment

5.1 Quality Assurance & Verification Benchmarks

FINSTΛNT underwent rigorous end-to-end testing across four verification dimensions: (1) **Mathematical Verification:** Multi-sheet test files with ground-truth financial figures were tested; calculations matched ground truth with 100% precision. (2) **Performance Benchmarking:** End-to-end response times averaged 3.2 to 4.5 seconds under standard workloads. (3) **Lifecycle Resilience:** Replacing and deleting live datasets was validated to confirm clean database table drops and metadata synchronization with zero orphan records. (4) **Session Resilience:** JWT token refresh pipelines and clock-skew leeway buffers were verified to prevent accidental logouts.

5.2 Security Architecture & Guardrails

Information security and confidentiality are baked directly into the system architecture:

- **AST SQL Security Guard:** All synthesized queries pass through an Abstract Syntax Tree inspection filter. Only read-only SELECT statements are permitted; destructive commands (DROP, DELETE, UPDATE, INSERT, ALTER) are automatically blocked.
- **Dual-Database Physical Isolation:** Application authentication data, user credentials, and session logs (app.db) are kept physically separate from the financial warehouse tables (live_data.db).
- **Zero Public AI Training:** Financial records are never submitted to public web chatbots or used to train external language models. Data processed via secure API endpoints adheres to strict zero-retention policies.
- **Principle of Least Privilege:** Users only receive the direct answer, table, or chart required to satisfy their query. Full master workbooks are never transmitted across user sessions.

5.3 Production Deployment & Hosting

The platform is deployed in a secure, centralized cloud environment on AWS Ubuntu EC2. Production availability is managed via PM2 process supervisors ensuring automated service restarts upon exceptions, paired with Nginx reverse proxying for HTTPS and SSL termination. The system maintains an ultra-lean memory footprint (~230 MB backend, ~60 MB frontend), enabling sustained production performance for just $5 to $8 per month.

6. Acceptance Criteria

The following formal criteria were defined for final project sign-off. All criteria have been thoroughly tested, validated, and approved:

Ref ID	Criterion	Validation Standard	Result
AC-01	Natural Language Querying	System accurately interprets natural English questions without requiring technical knowledge.	**Passed**
AC-02	Mathematical Correctness	All calculations match ground-truth spreadsheet aggregations with 100% computational accuracy.	**Passed**
AC-03	Turnaround Latency	End-to-end response times remain under 5 seconds for standard analytical requests.	**Passed**
AC-04	Custom Layout Support	System dynamically ingests non-standard spreadsheets with diverse column layouts and formulas.	**Passed**
AC-05	Multi-Device Remote Access	Users can access the system securely from any authorized mobile, tablet, or desktop browser.	**Passed**
AC-06	Role-Based Permissions	Access to datasets is strictly governed by admin permissions; standard users only see approved data.	**Passed**
AC-07	Live Dataset Lifecycle	Admins can replace and delete active datasets cleanly in real-time with full dynamic table purge.	**Passed**
AC-08	Visualizations & Export	System automatically renders interactive charts and supports instant one-click PDF/CSV export.	**Passed**
AC-09	Cost Optimization	Monthly cloud hosting and API usage costs do not exceed the target threshold of $5 to $8 / month.	**Passed**

7. Conclusion

FINSTΛNT successfully fulfills all strategic objectives established at project commencement. It replaces rigid off-the-shelf software with an adaptive intelligence engine, breaks the physical barrier of desktop-bound spreadsheets, removes person-to-person administrative delays, and provides granular role-based security that eliminates master workbook over-sharing.

By combining natural language understanding with deterministic database execution, FINSTΛNT delivers verified, mathematical financial answers in seconds, empowering decision-makers with immediate operational clarity anytime, anywhere—at an unbeatable cost of $5 to $8 per month.

The codebase, database architectures, administrative interfaces, and production deployments are fully verified, operational, and officially signed off for production delivery.